ME302: 2025-26-II
Course Project Report

Compressible Flow Testing: Maximum Model Scale Analysis

Priyank Agnihotri (230797) | Shivansh Gupta (230976) | Toshika Kamble (231099)

1 Problem Description

This project involves the scaling analysis of a turbomachine component, tested in a closed-loop high-pressure facility. The flow is driven by a compressor ($1 \rightarrow 2$) and regulated by a throttle chamber ($4 \rightarrow 5$) to maintain constant inlet stagnation conditions. The goal is to determine the maximum geometric scale $s = l_{model}/l_{proto}$ that satisfies dynamic similarity and facility constraints given below:

- **Similarity:** Mach number $M = 0.5$ and Reynolds number $Re = 3 \times 10^6$
- **Minimum Static Pressure:** $p_{min} \geq 200$ kPa (structural requirement)
- **Pressure Balance:** Closed-loop requirement $p_{05} \geq p_{01}$
- **Losses due to pressure:** $p_{05} = 0.95p_{04}$ and $p_{03} = 0.985p_{02}$
- **Inlet Stagnation Temperature:** $T_{01} = 293$ K
- **Geometric:** $l_{proto} = 0.2$ m, $D_3 = 0.6$ m, $L = 1.5$ m, and $D_4 = 2l_{model}$

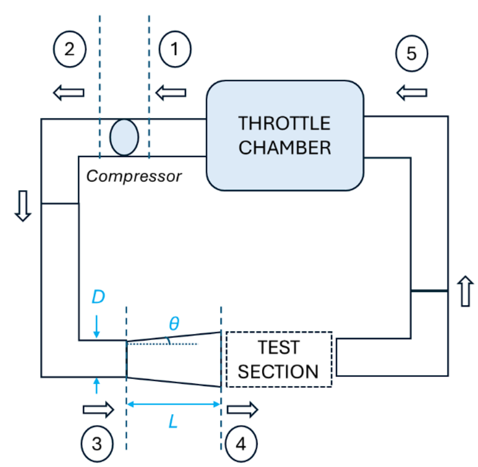

Figure 1: Compressor characteristic curve with selected operating point

2 Methodology

2.1 Isentropic Relations at the Test Section

The flow properties at the test section are obtained using isentropic compressible flow relations. For a Mach number $M_4 = 0.5$ and ratio of specific heats $\gamma = 1.4$, the parameter ψ is defined as:

$$\psi = 1 + \frac{\gamma - 1}{2} M^2 = 1.05$$

Using this parameter, the pressure and temperature ratios at station 4 are:

$$\frac{p_4}{p_{04}} = \psi^{-3.5} = 0.843, \qquad \frac{T_4}{T_{04}} = \frac{1}{\psi} = 0.952$$

2.2 Test Section Geometry

The characteristic dimensions are:

$$l_{\text{model}} = 0.2s, \qquad D_4 = 2l_{\text{model}} = 0.4s$$

where s denotes the scale factor of the model. Since $D_4 < D_3 = 0.6\,\text{m}$, the section between stations 3 and 4 is converging. Therefore, the loss coefficient, K is assumed to be zero, and the stagnation pressures satisfy

$$p_{04} = p_{03}.$$

2.3 Reynolds Number Matching

To maintain dynamic similarity, the Reynolds number is matched. Using compressible flow relations, the required static pressure at station 4 is

$$p_4 = \frac{Re \cdot \mu}{l_{\text{model}} M_4 \sqrt{\gamma/(RT_4)}}$$

The corresponding stagnation pressure is then given by

$$p_{04} = \frac{p_4}{0.843}.$$

2.4 Facility Pressure Chain

The stagnation pressure relations across the facility loop are

$$p_{03} = p_{04}$$
$$p_{02} = \frac{p_{04}}{0.985}$$
$$p_{01} = \frac{p_{02}}{PR}$$
$$p_{05} = 0.95\, p_{04}$$

For stable closed-loop operation, the condition

$$p_{05} \geq p_{01}$$

must hold, which leads to the requirement

$$PR \geq 1.0687.$$

2.5 Computational Method

The computational analysis was carried out using Python, with NumPy, pandas, and Matplotlib used for data handling and visualization. The compressor map dataset was handled through a structured workflow to examine all operating conditions consistently.

A point-by-point analysis was performed, where each operating condition was checked against the governing constraints. The corresponding model scale was computed for all valid points, and the maximum feasible value was tracked to identify the optimal operating condition.

3 Results and Discussion

The plot in Figure 2 shows the compressor characteristic curve with the selected operating point marked by the green star. This point is located at the high-flow limit of the performance map, indicating that the system is mass-flow limited. Since the operating point is at the edge of the characteristic curve, it verifies that the compressor's maximum flow capacity is the primary constraint on the model scale.

The minimum structural pressure limit (200 kPa) and the closed-loop feasibility constraint are both satisfied well within the limit. Specifically, the static pressure in the test section reaches 212.60 kPa, and the return pressure downstream ($p_{05} = 239.58$ kPa) is above the required compressor inlet pressure ($p_{01} = 211.51$ kPa).

The maximum corrected mass flow is primary limiting factor for the model size which is present on the compressor map. Since the required mass flow rate is proportional to the test section area (s^2), an increase in the geometric scale the flow demand will reach the compressor's choking.

Final Design Parameters

Geometric Scale

- Maximum Feasible Scale: $s = 0.6384$
- Model Length: $l_{\text{model}} = 127.7$ mm
- Test Section Diameter: $D_4 = 255.4$ mm

Compressor Operating Point

- Pressure Ratio (p_{02}/p_{01}): 1.2105

- Corrected Mass Flow ($\dot{m}_{\text{ref}}$): 10.5503 kg/s
- Temperature Ratio (T_{02}/T_{01}): 1.0685

Nodal Pressures

Station	Pressure (kPa)
p_{01} (Inlet)	211.51
p_{02} (Exit)	256.03
p_{03}	252.19
p_{04}	252.19
p_{05}	239.58
p_4 (Static)	212.60

Table 1: Calculated nodal pressures at the maximum scale

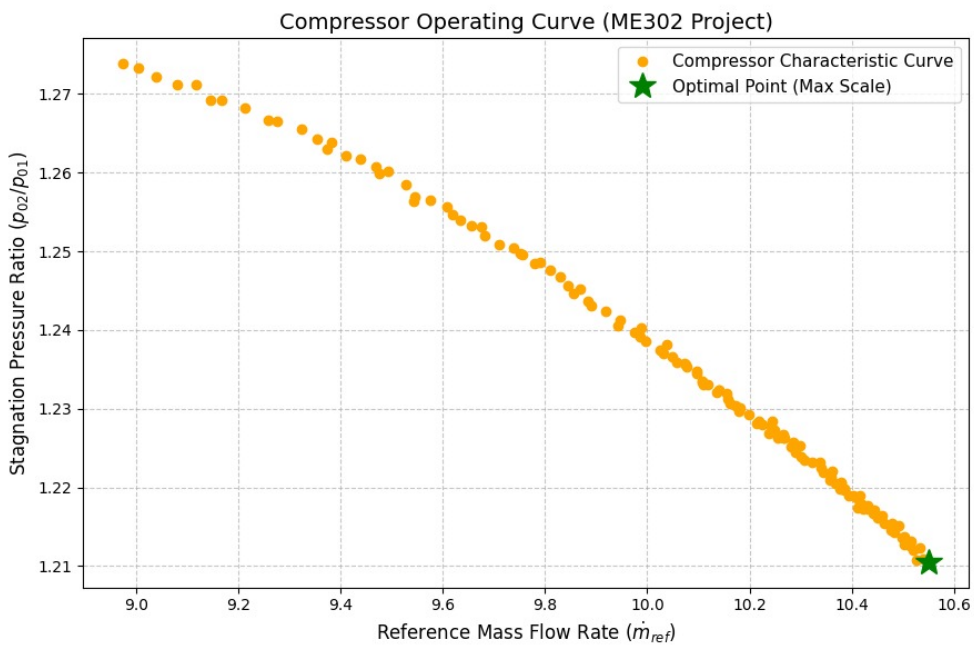

Figure 2: Compressor characteristic curve with selected operating point

4 Supplement

Files uploaded in addition to the report :

- "me302.py" - Python script used for calculations
- "compressor_map.xlsx" - Compressor map dataset